

Ethanol Blending in Gasoline – Vietnam

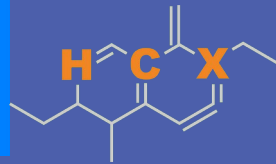
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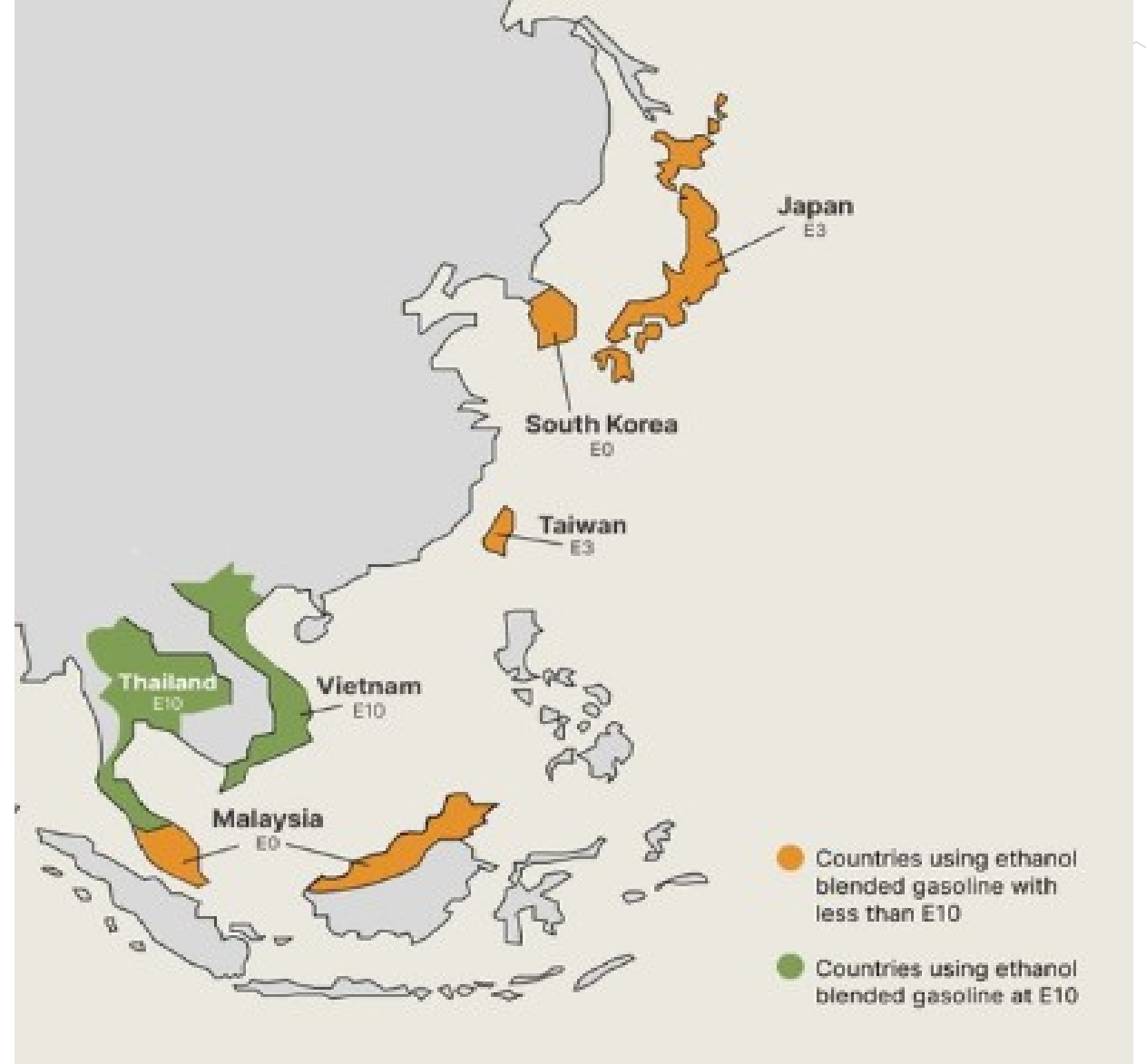


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Ethanol Blending in Asia

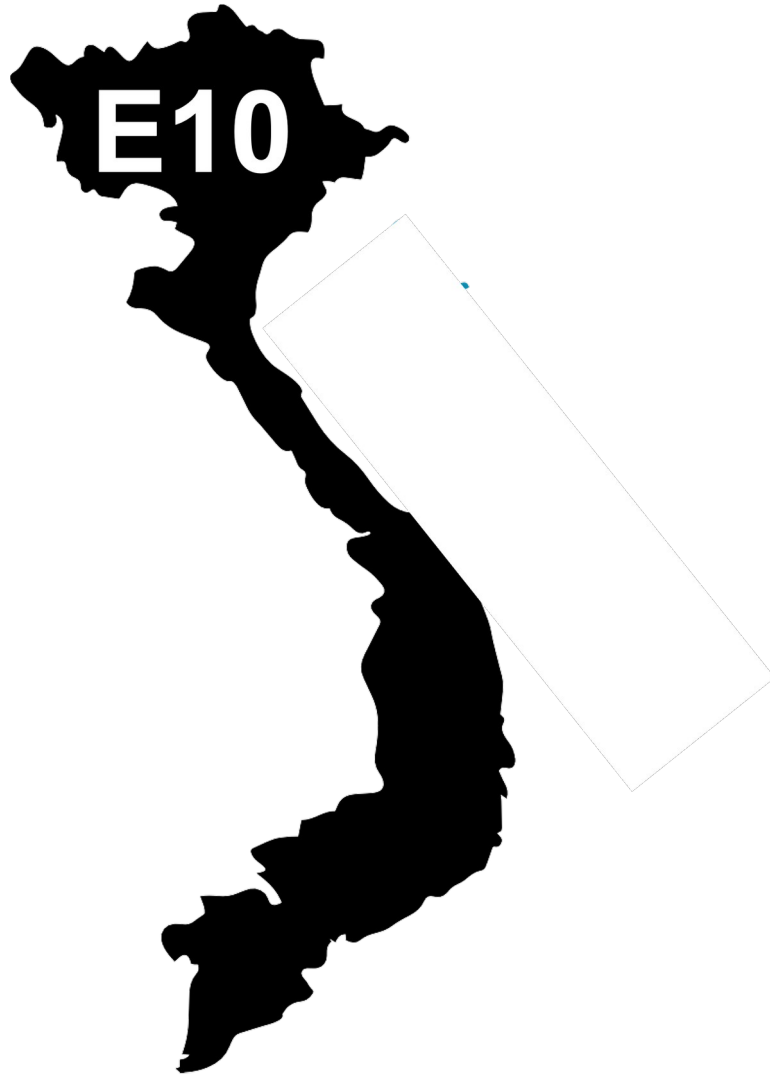
There are important fuel quality and environmental impact of vehicle emission challenges in these regions.

- The use of ethanol improves gasoline quality and creates flexibility in gasoline production.
- Ethanol use is a cost-effective way to increase gasoline octane and to replace more expensive gasoline components.
- Ethanol contributes to transport decarbonization and air quality improvement.
- There are opportunities across Asia to increase the ethanol blend level and implement new policies on the use of gasoline-ethanol blends.



Seven countries with potential and additional use of ethanol were studied: 1) Gasoline market profiles; 2) Optimization of gasoline blends with ethanol and, 3) Environmental impact of gasolines blended with ethanol.

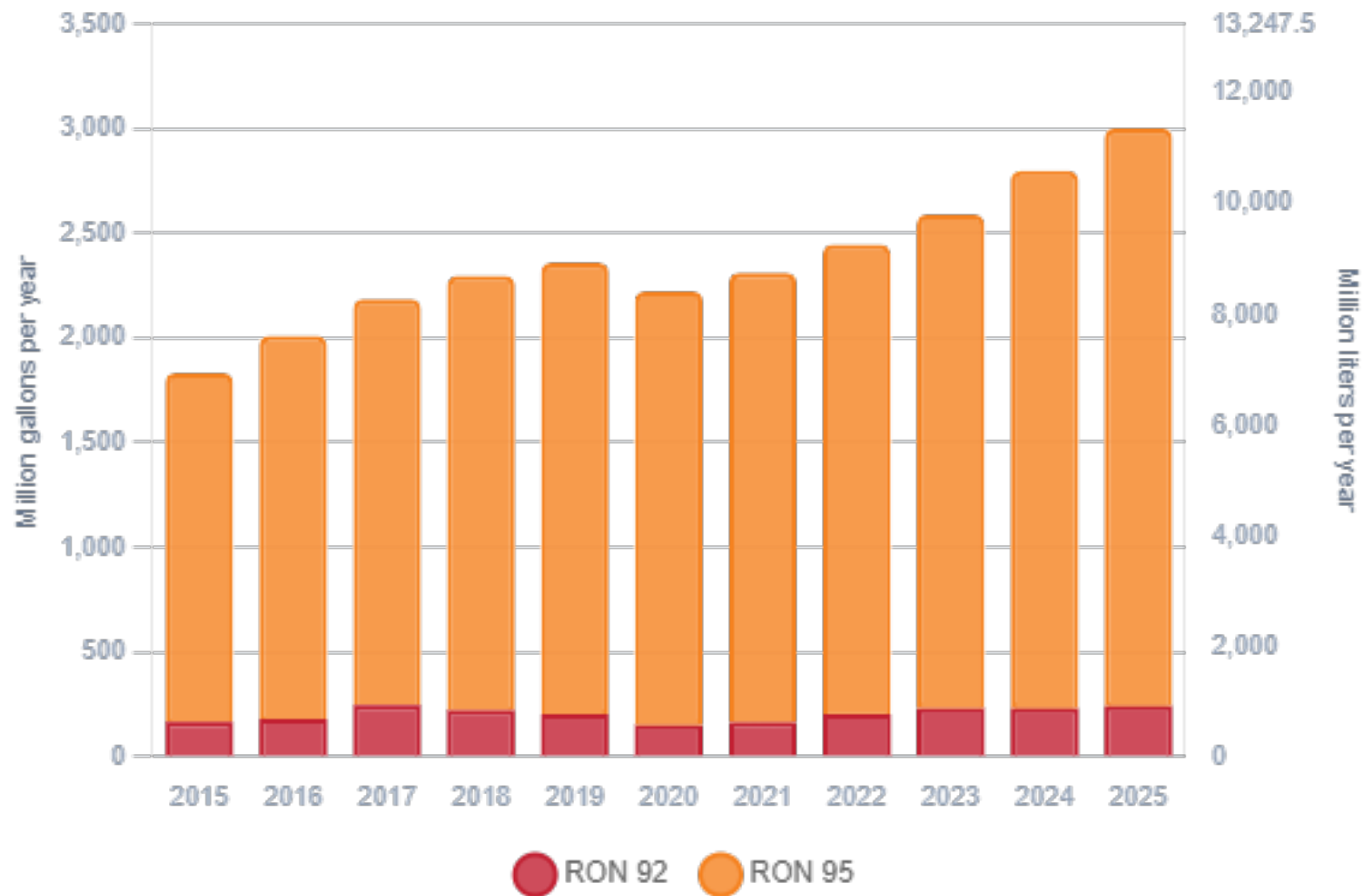
Ethanol Gasoline Blending in Vietnam



In 2025, gasoline consumption in Vietnam was 11,373 million liters (3,005 million gallons) and growth rate was 3.3%. Gasoline production and net imports were 7,949 and 3,423 million liters (2,100 and 904 million gallons) correspondingly. Vietnam complies offers three grades: RON 92 and RON 95 with an average octane of 94.7 RON.

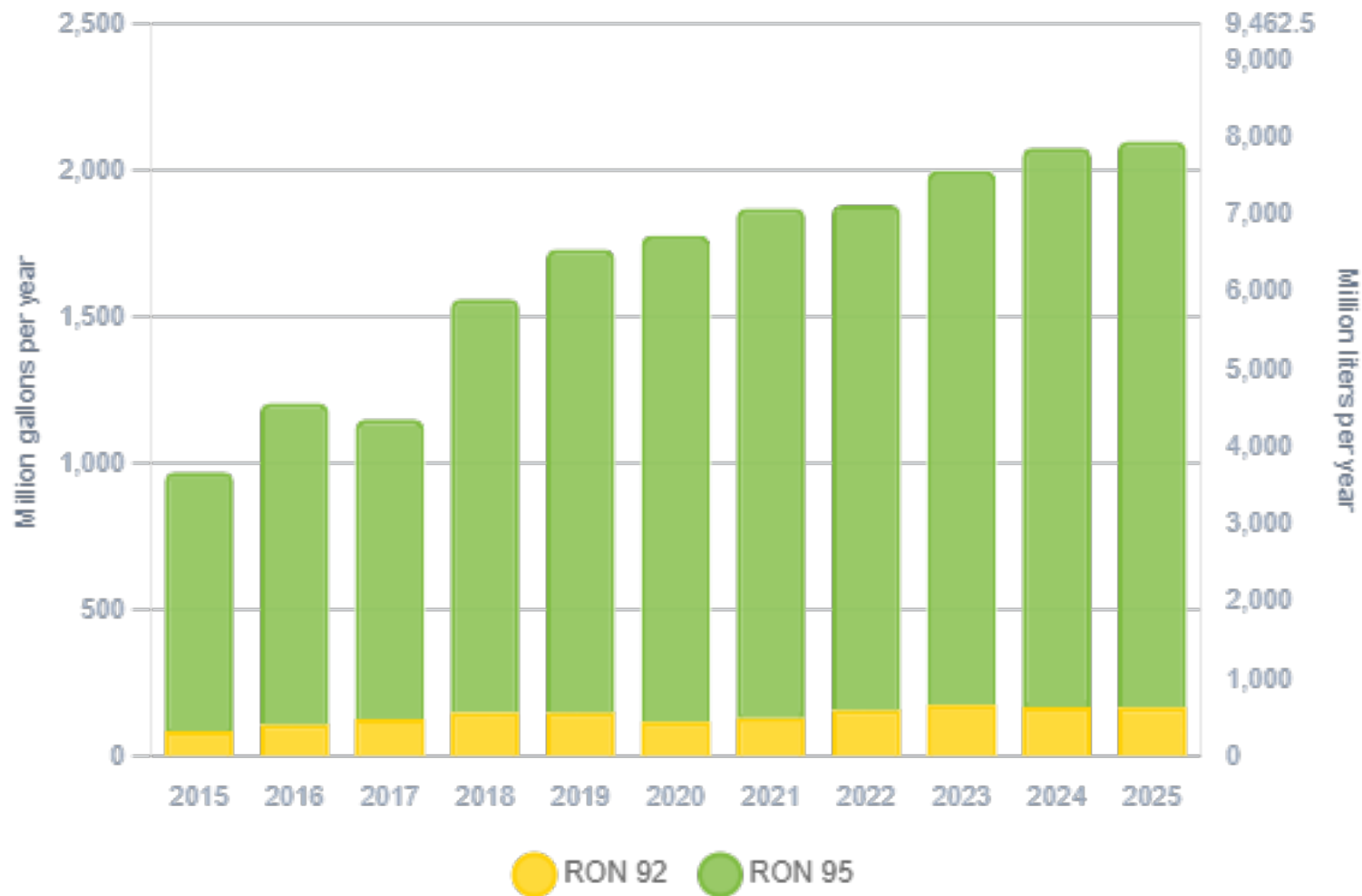
Vietnam's bioethanol demand is growing rapidly, driven by government policies like a planned nationwide E10 gasoline mandate set to begin in June 2026. Ethanol consumption in Vietnam was 369 million liters (97 million gallons) representing 3.2% of gasoline demand. Ethanol production and Net Imports were 250 and 119 million liters (66 and 31 million gallons) correspondingly.

Gasoline Demand in Vietnam



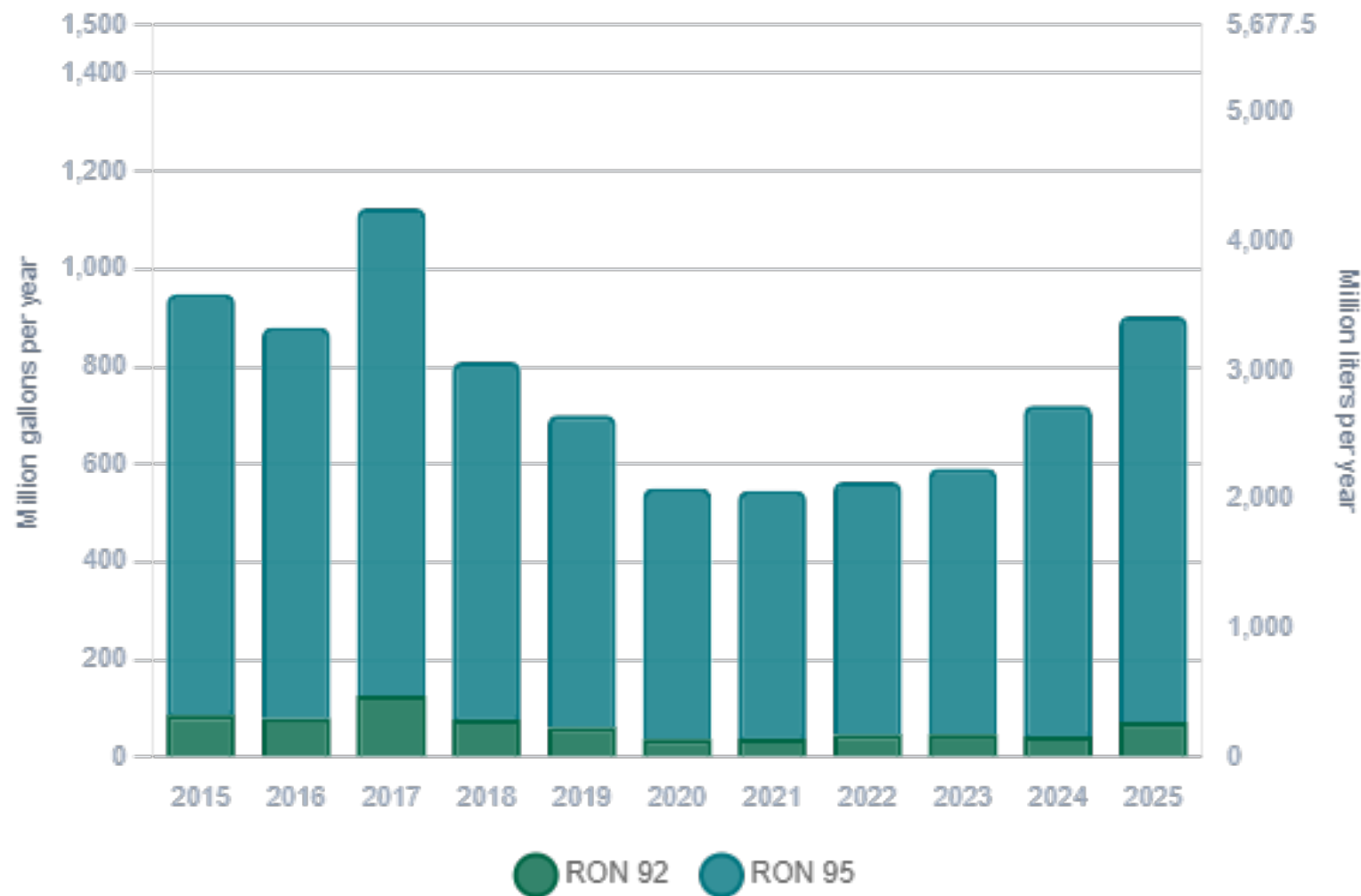
Source: Petro Vietnam, EIA, EIT data

Gasoline Production in Vietnam



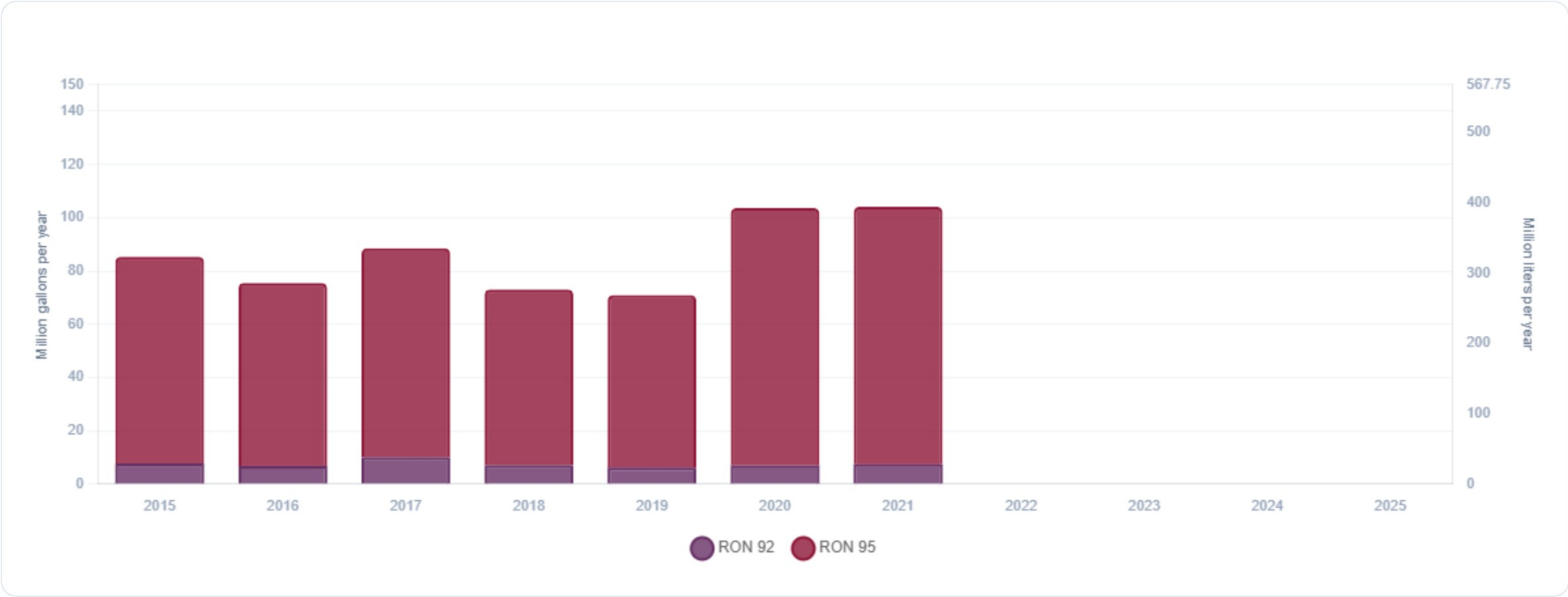
Source: Petro Vietnam, EIA, EIT data

Gasoline Imports in Vietnam



Source: Petro Vietnam, EIA, EIT data

Gasoline Exports in Vietnam



Source: Petro Vietnam, EIA, EIT data

Ethanol Balance in Vietnam



Source: Petro Vietnam, USDA

Gasoline Quality in Vietnam

Name	QCVN 01:2022/BKHCN					
Implementation Year	2023					
Applicability	All territory					
Grade	RON 92		RON 95		RON 97	
	Min	Max	Min	Max	Min	Max
RON	92.0		95.0		97.0	
MON	-		-		-	
AKI	-		-		-	
Benzene (%vol)		1.0		1.0		1.0
Aromatics (% vol)		35.0		35.0		35.0
Olefins (%vol)		18.0		18.0		18.0
Lead (g/L)		0.005		0.005		0.005
Manganese (mg/L)		5.0		5.0		5.0
Sulfur (ppm)		10		10		10
Oxygen (%w)		2.7		2.7		2.7
Ethanol (%vol)		-		-		-
MTBE (%vol)		-		-		-
RVP (psi)		8.7		8.7		8.7

Source: Caselaw

The FARO90 logo is shown in a blue box, and a chemical structure is displayed to its right.

Catalytic Gasoline
Light Primary Naphta
Heavy Primary Naphta
Reformate
Isopentane
Alkylate
Isomerate
Hydrocracked Gasoline
Coker Naphta
n-Butane
MTBE
ETBE
TAME
Aromatics



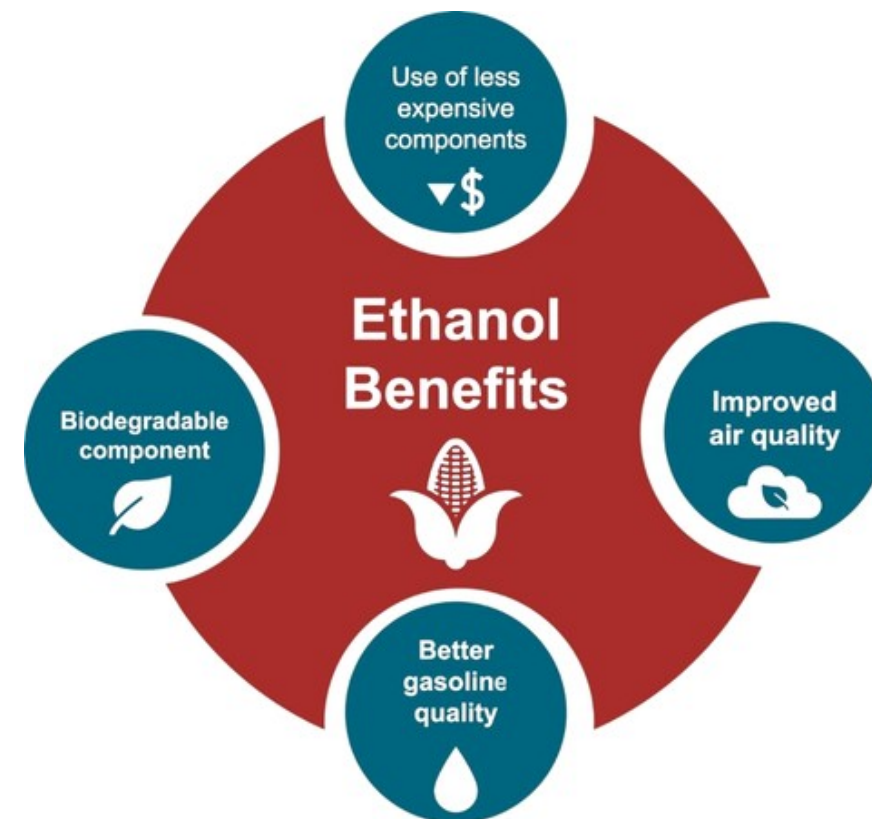
Gasoline Blending Optimization

In some parts of the world, ethanol is added to gasoline as a blending component. The advantages of ethanol include that it is a renewable fuel made of biomass; that it is an octane booster that helps to dilute sulfur; and that it allows the fulfillment of environmental objectives. To determine the optimal components to be blended with ethanol, a **blending model** was used. This model selects the components to add in the gasoline/ethanol blend based on:

- Components prices,
- Properties each component affects,
- Quality parameters by country, and
- Component availability by country.

Through iterations, the model obtains the %v/v of the components to be blended with 10%, 15%, 20%, 25% and 30% of ethanol, in such a way that the final blend complies with the required properties of a finished gasoline by country.

The blending model uses gasoline component spot average prices 2025 and provides fuel prices that do not include country distribution costs, local taxes and subsidies and import or gas station margins.

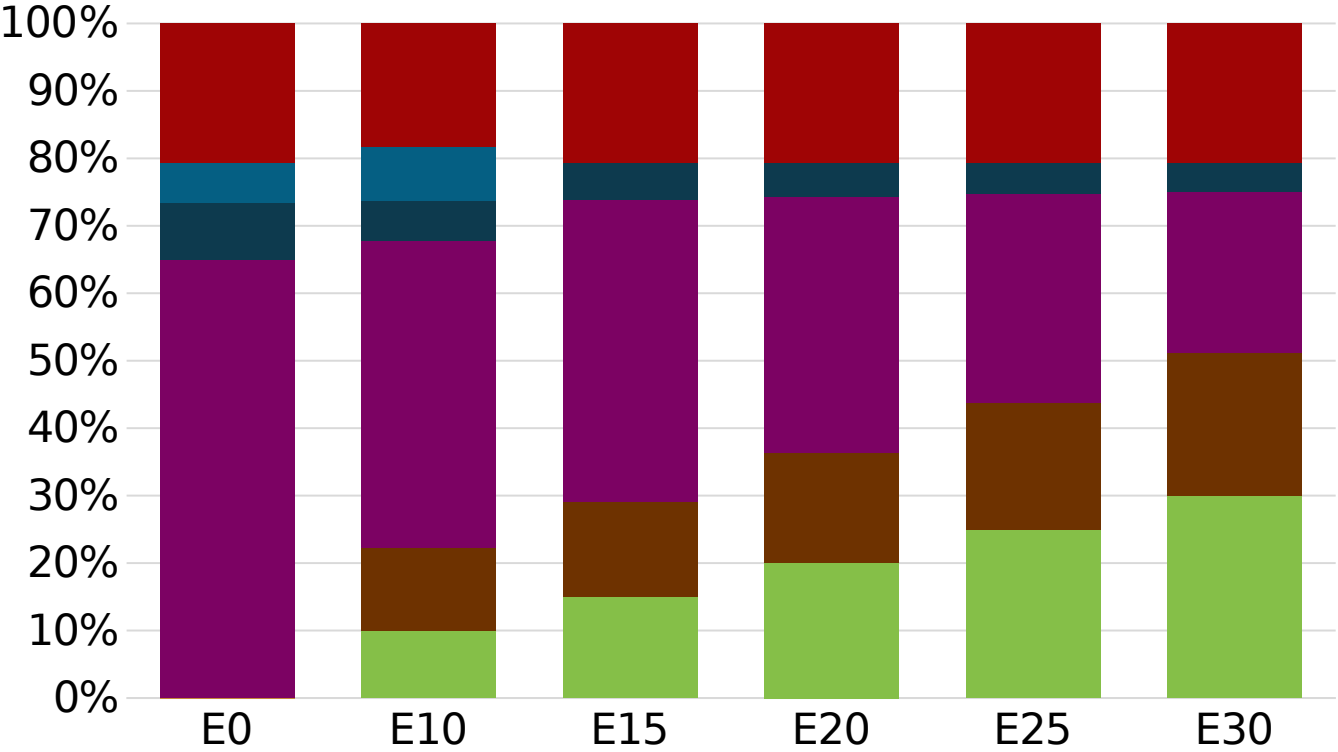


Vietnam Available Blending Components



Source:
Faro90

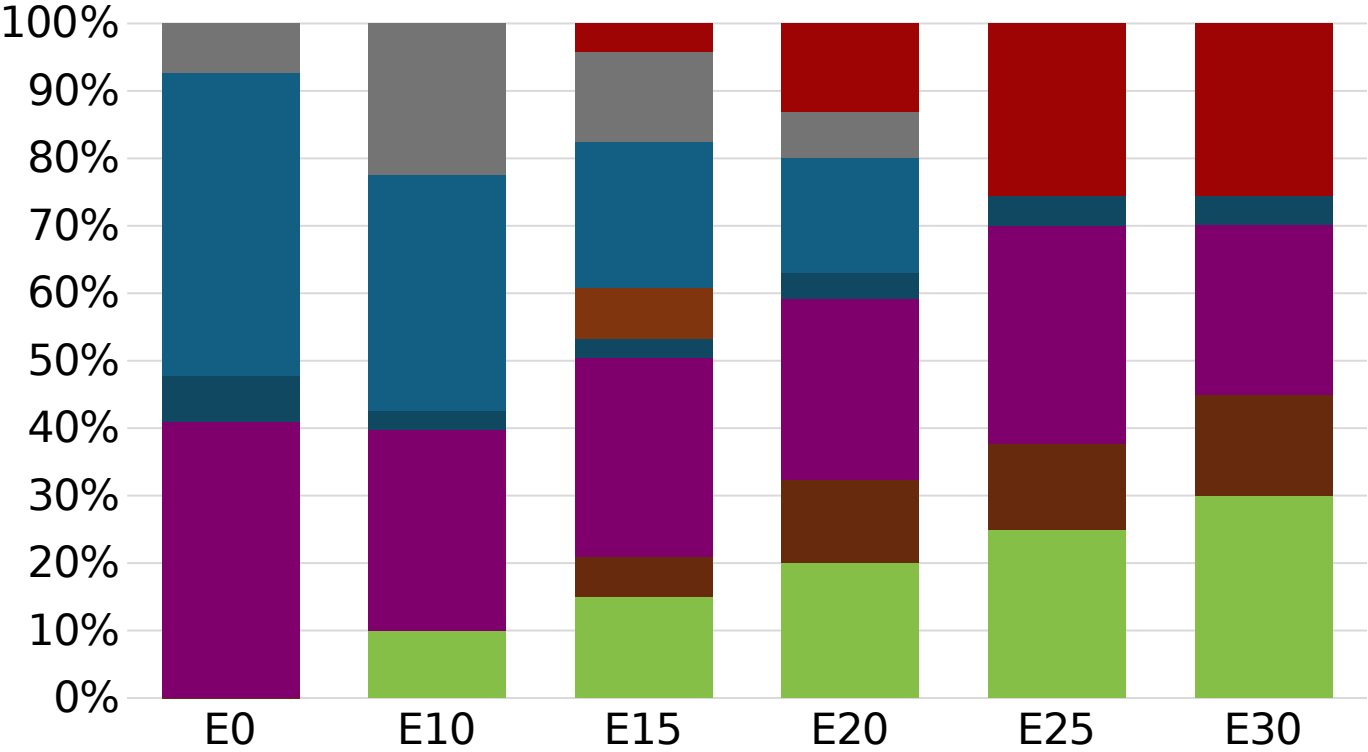
Vietnam – Gasoline 92 – Constant Octane



Average CIF 2025
Prices

Octane (RON)	92.0	92.0	92.9	94.0	95.0	96.0
Price (US\$/gal)	\$ 2.067	\$ 1.900	\$ 1.850	\$ 1.807	\$ 1.759	\$ 1.707

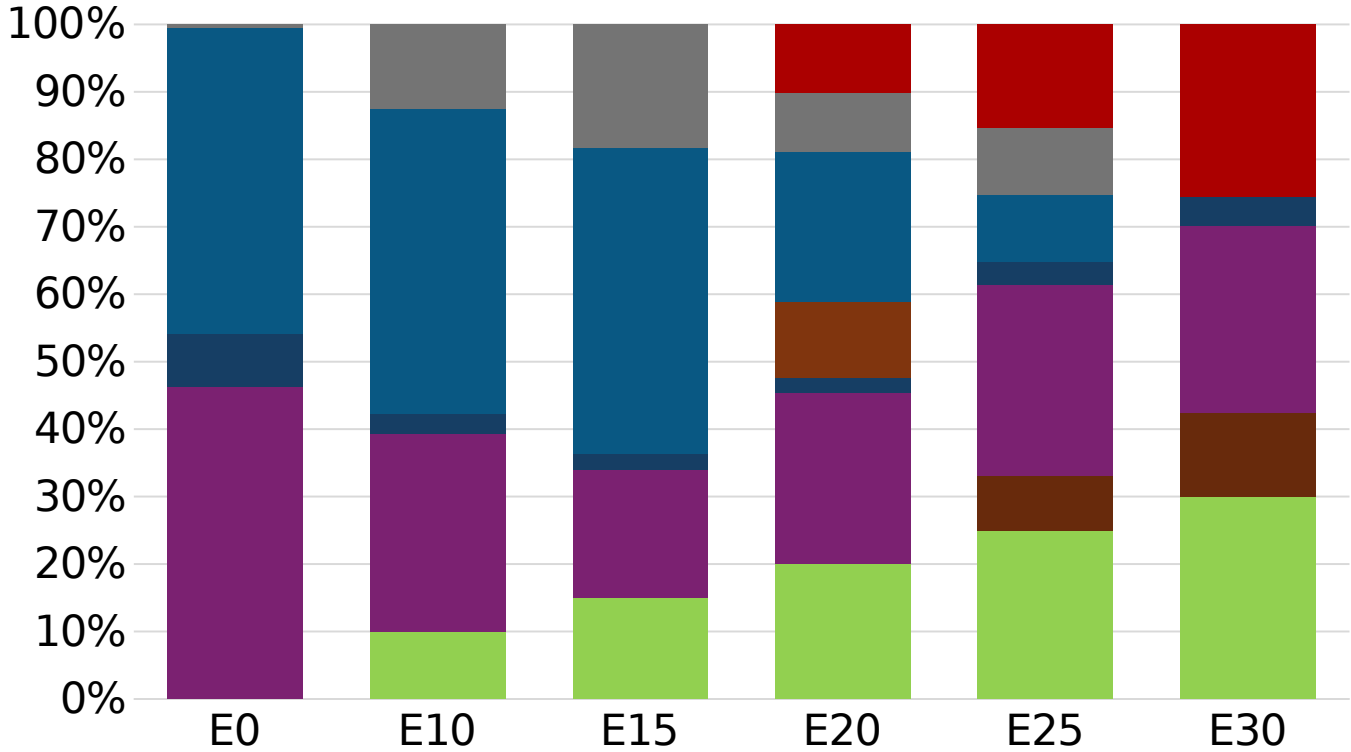
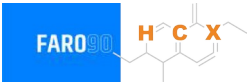
Vietnam – Gasoline 95 – Constant Octane



Average CIF 2025
Prices

Octane (RON)	95.0	95.0	95.0	95.0	95.2	96.3
Price (US\$/gal)	\$ 2.223	\$ 2.092	\$ 2.001	\$ 1.877	\$ 1.768	\$ 1.720

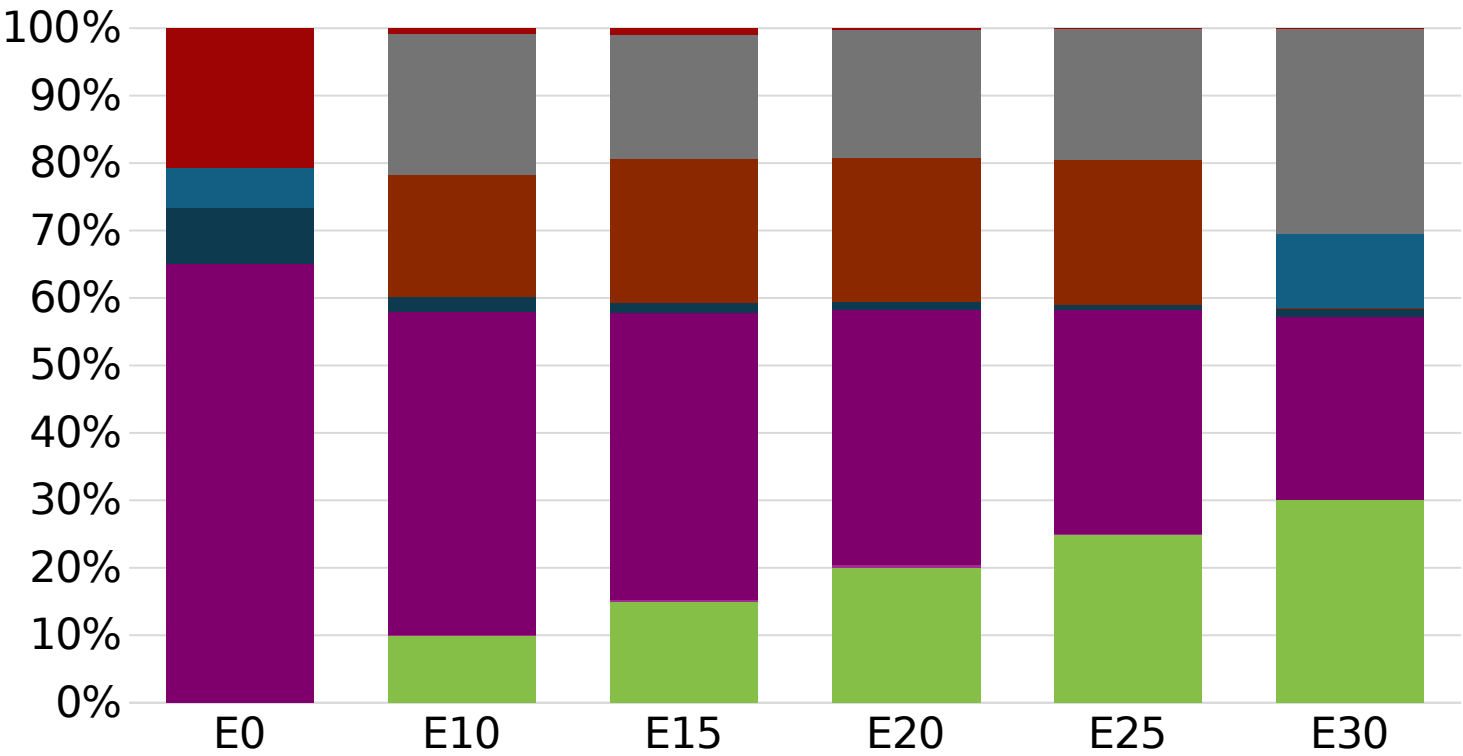
Vietnam - Gasoline 97 - Constant Octane



Average CIF 2025
Prices

Octane (RON)	97.0	97.0	97.0	97.0	97.0	97.0
Price (US\$/gal)	\$ 2.329	\$ 2.203	\$ 2.108	\$ 2.015	\$ 1.892	\$ 1.770

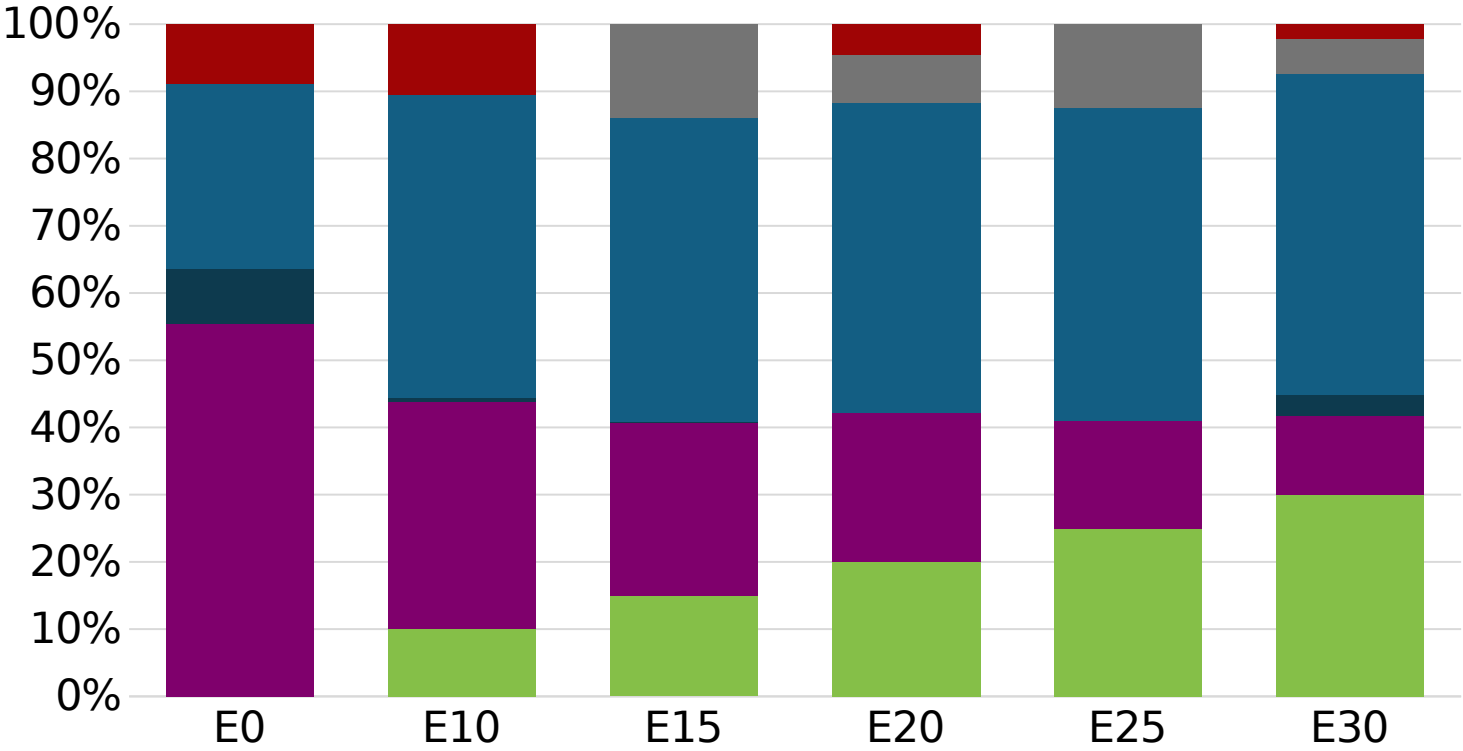
Vietnam – Gasoline 92 – Increased Octane



Average CIF 2025
Prices

Octane (RON)	92.0	94.1	95.7	97.4	99.0	101.0
Price (US\$/gal)	\$ 2.067	\$ 2.067	\$ 2.067	\$ 2.067	\$ 2.067	\$ 2.067

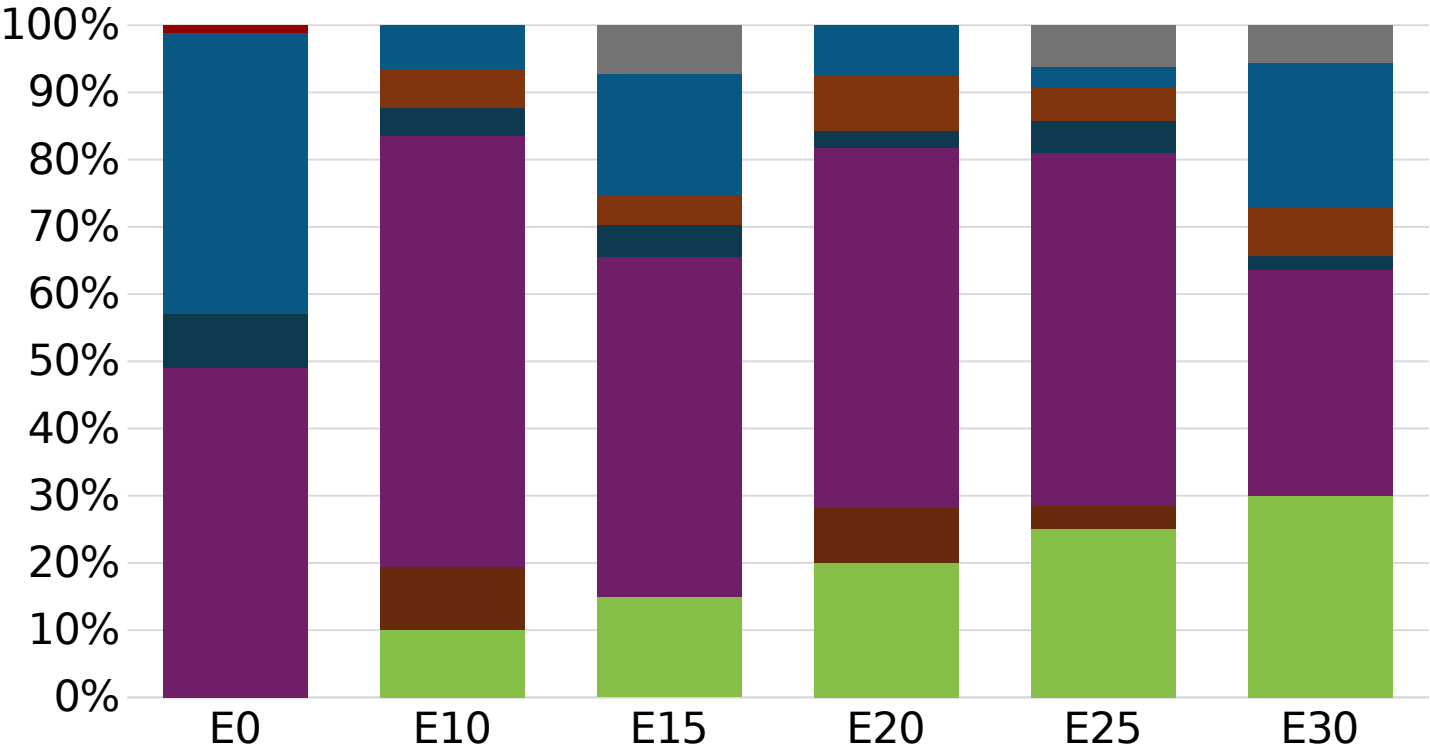
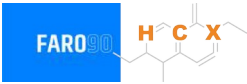
Vietnam – Gasoline 95 – Increased Octane



Average CIF 2025
Prices

Octane (RON)	95.0	96.8	98.4	100.0	101.6	103.7
Price (US\$/gal)	\$ 2.223	\$ 2.223	\$ 2.223	\$ 2.223	\$ 2.223	\$ 2.223

Vietnam – Gasoline 97 – Increased Octane



Average CIF 2025
Prices

Octane (RON)	97.0	99.1	101.1	101.9	103.9	104.9
Price (US\$/gal)	\$ 2.327	\$ 2.327	\$ 2.327	\$ 2.327	\$ 2.327	\$ 2.327

Vehicle Emission Impact for Ethanol Gasoline Blending

The model used in this analysis takes as a reference the [International Vehicle Emissions Model \(IVE\)](#).

The model uses the Base Emission Rates from IVE model, as well as its Adjustment Factors based on:

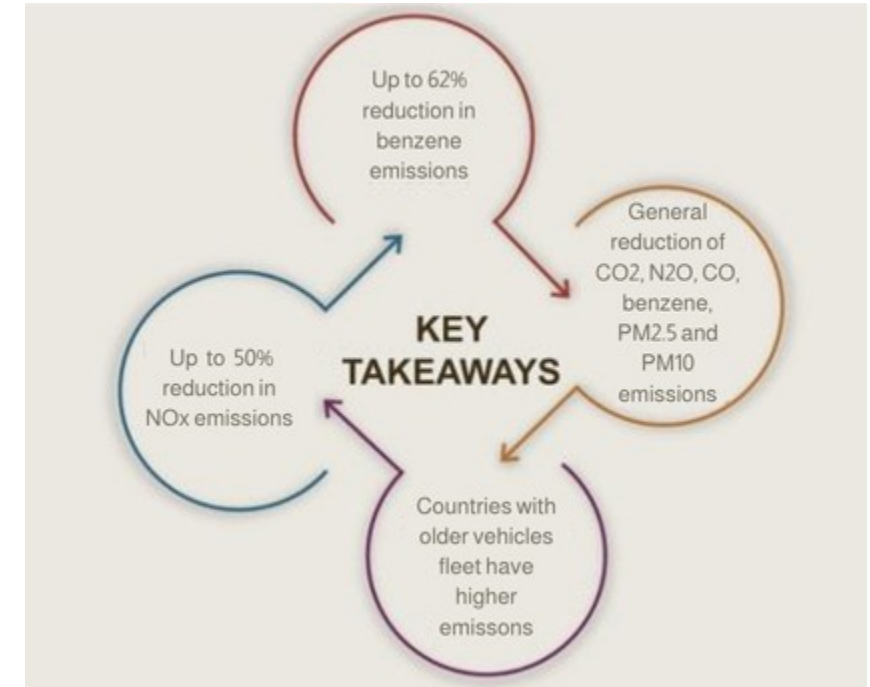
- Vehicle technology (cars, trucks, buses, motorcycles),
- Vehicle fleet average age,
- Average traveled distance per vehicle by country, as well as
- Geographical and climatic conditions (altitude, humidity, temperature).

Emissions of criteria pollutants, toxic pollutants, and greenhouse gases (GHG) were calculated and calibrated with emission inventories, using real gasoline quality data. The reduction rates for gasoline/ethanol blends were obtained from various sources (IPCC, US Grains, among others).

Emission estimations for different pollutants for gasoline and gasoline/ethanol blends (10%, 15%, 20%, 25% and 30% ethanol) were determined using the IVE Model. A comparison between the results and the European (Euro 6) requirements is made. Results are also compared with real emissions of the United States vehicle fleet*.

*Source: Bureau of transportation statistics.

Main Results



Gasoline Vehicle Fleet - Vietnam

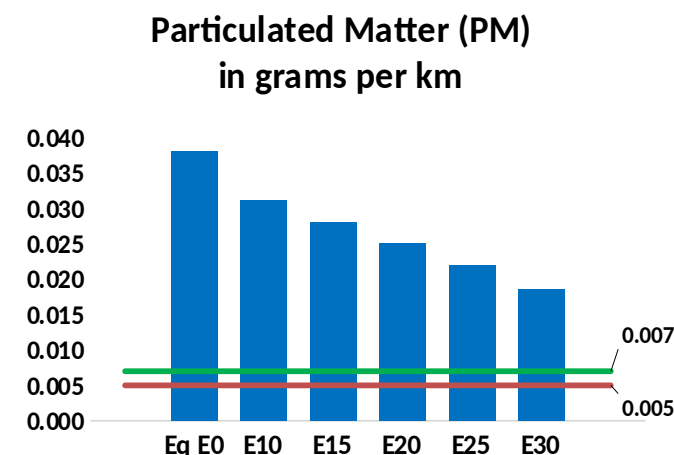
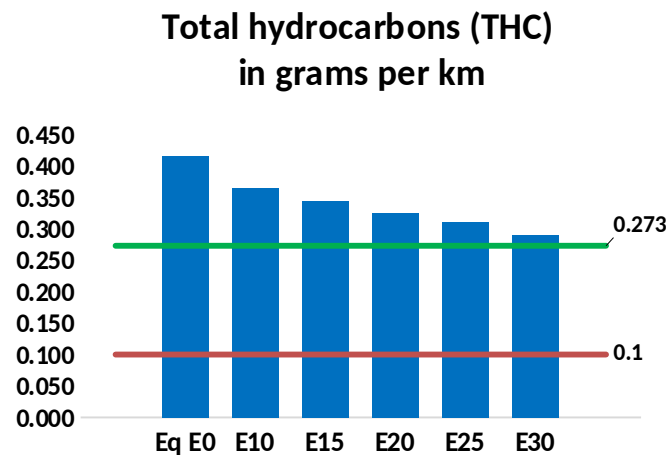
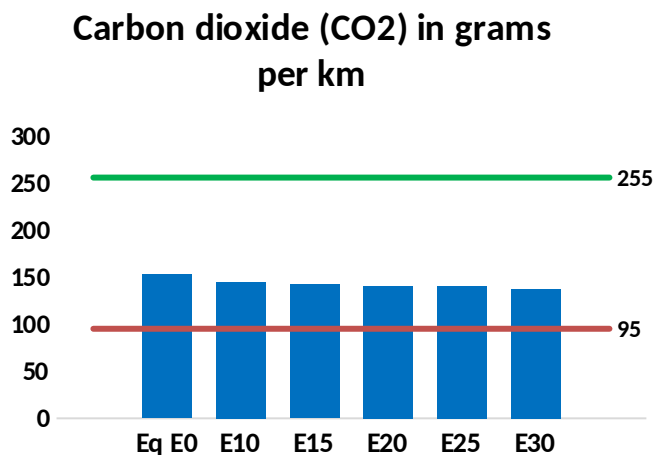
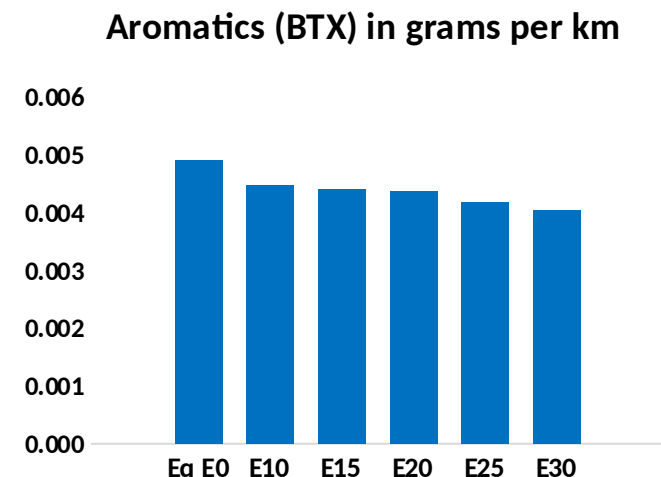
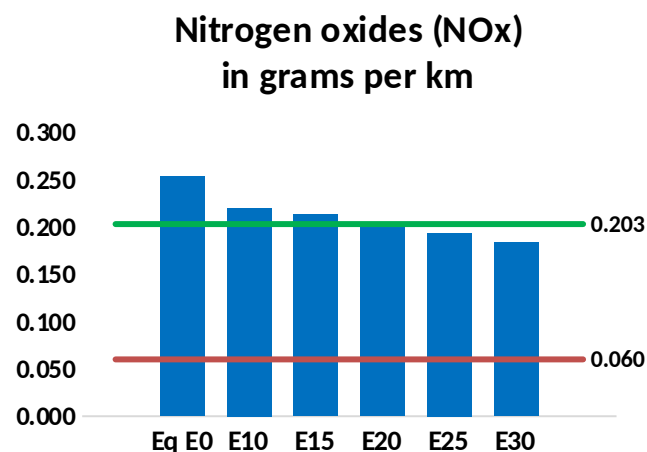
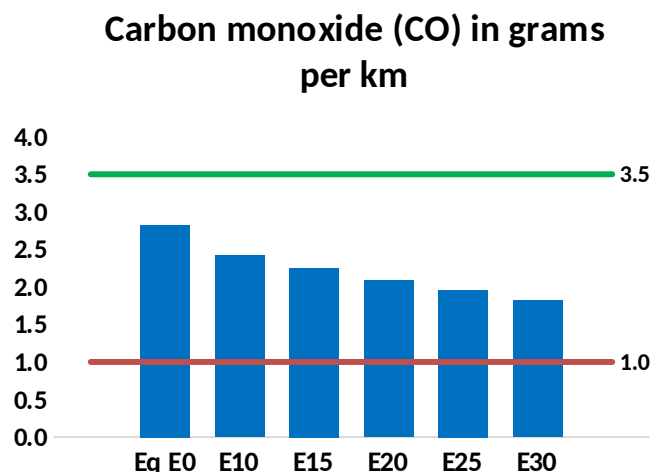
Type	Weight	Engine	Exhaust	Recovery	Age	No. Vehicles				
Auto/Sml Truck	Light	Carburator	3-Ways	PCV Tank	>8 years	611				
		Single FI	3-Ways		>8 years	18,721				
		Multi FI	3-Ways		>8 years	75,441				
			Euro III		>8 years	96,476				
			Euro IV		>8 years	298,553				
			Euro V		>8 years	198,570				
			Euro 6		>8 years	469,620				
					4-8 years	590,965				
		<4 years	660,304							
		Hybrid	All		41,100					
	Electric			All	123,300					
	Medium	Carburator	3-Ways	PCV Tank	>8 years	281				
		Single FI	3-Ways		>8 years	8,602				
		Multi FI	3-Ways		>8 years	34,664				
			Euro III		>8 years	44,329				
			Euro IV		>8 years	137,179				
			Euro V		>8 years	91,239				
			Euro 6		>8 years	215,780				
					4-8 years	271,535				
		<4 years	303,396							
		Hybrid	All		18,900					
		Electric			All	56,700				
Sml Engines		Light	4-Cycle		None	PCV	>8 years	1,449,187		
	Euro III			>8 years	1,475,228					
	Euro IV			>8 years	4,377,632					
	Euro V			>8 years	2,727,054					
	Euro 6			>8 years	7,854,549					
				4-8 years	19,522,119					
	<4 years			7,715,447						
	Hybrid		All	0						
	Electric			All	2,300,000					
Truck	Heavy	Carburator	3-Ways	PCV	>8 years	530				
		Fuel-Injection	3-Ways		>8 years	119,365				
			Euro III		>8 years	131,538				
			Euro IV		>8 years	417,544				
			Euro V		>8 years	245,806				
			Euro 6		>8 years	665,533				
					4-8 years	837,500				
			<4 years		935,767					
		Hybrid	All		0					
		Electric			All	1,534				

Vehicle Fleet: 54,532,599
Gasoline Vehicle Fleet: 41,162,034

Source: MoIT

Vietnam – Vehicular Emissions

TIER III BIN 125 United States
Euro 6 Standard



Vietnam – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	2,141,325	1,986,421	1,831,517	1,706,474	1,581,513	1,485,689	1,382,999	-14%	-26%
CO2	115,669,385	112,695,029	109,885,916	107,676,630	106,583,389	106,165,054	104,208,127	-5%	-8%
VOC	315,136	295,758	276,380	261,447	246,855	235,766	220,505	-12%	-22%
NOx	191,893	177,181	167,018	161,805	153,447	146,877	139,153	-13%	-20%
SOx	1,241	1,147	1,053	974	898	815	729	-15%	-28%
NH3	51,080	50,575	50,069	50,306	50,872	51,270	51,601	-2%	0%
N2O	1,830	1,820	1,812	1,821	1,859	1,879	1,901	-1%	2%
CH4	81,580	81,580	81,580	82,803	84,598	86,148	87,617	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	2,443	2,324	2,204	2,119	2,040	1,981	1,895	-10%	-17%
Acetaldehyde	6,540	7,506	11,013	16,772	22,850	26,111	30,853	68%	249%
Formaldehyde	25,725	25,010	29,155	34,234	35,865	39,677	43,192	13%	39%
Benzene	3,729	3,584	3,396	3,342	3,314	3,170	3,067	-9%	-11%
PM2.5	8,263	7,512	6,761	6,100	5,453	4,760	4,034	-18%	-34%
PM10	20,587	18,715	16,844	15,197	13,585	11,858	10,050	-18%	-34%

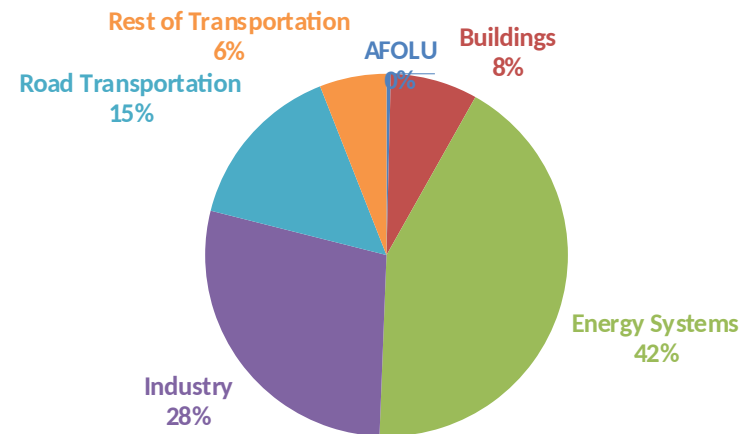
Life Cycle GHG Emissions

The Global Agenda for Climate Action calls for countries, cities and regions, businesses, and members of civil society around the world to achieve a low-carbon and climate-resilient economies in support of the Paris Agreement. Road Transportation accounts for 15% of total GHG emissions (IPCC,2023).

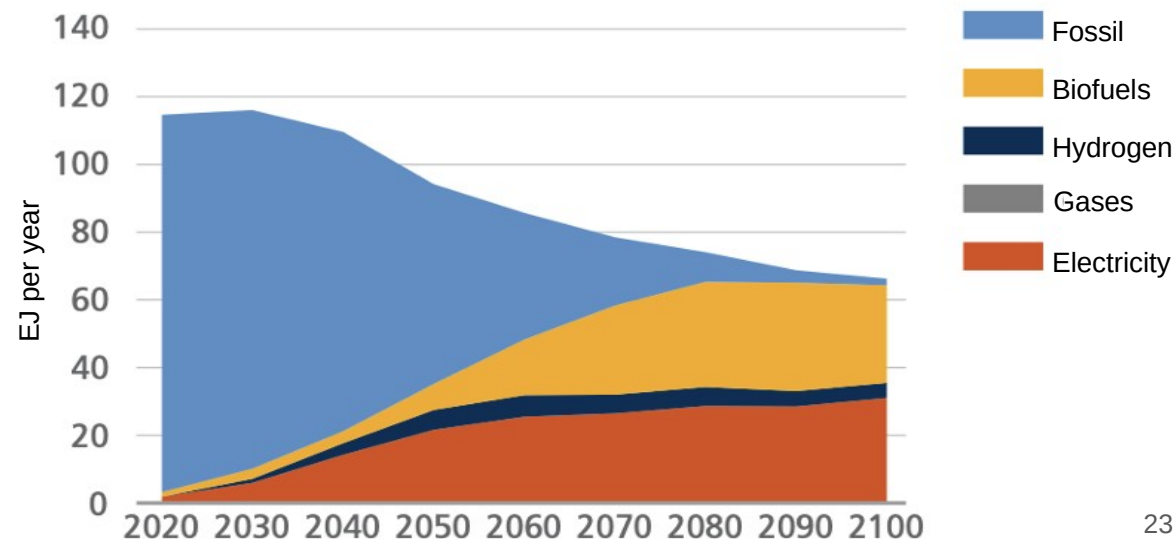
Ethanol gasoline blends are an internationally adopted practice that offer significant short-term benefits. Biofuels are an important solution to achieve reduction targets in the transportation sector.

Country GHG emission reductions are calculated using **RED II/III** and **GREET** life cycle emission methodologies and the **International Vehicular Emissions Model (IVE)**. Results are compared to the current country targets for 2035. Current 2025 country emission are those published by **IPCC**.

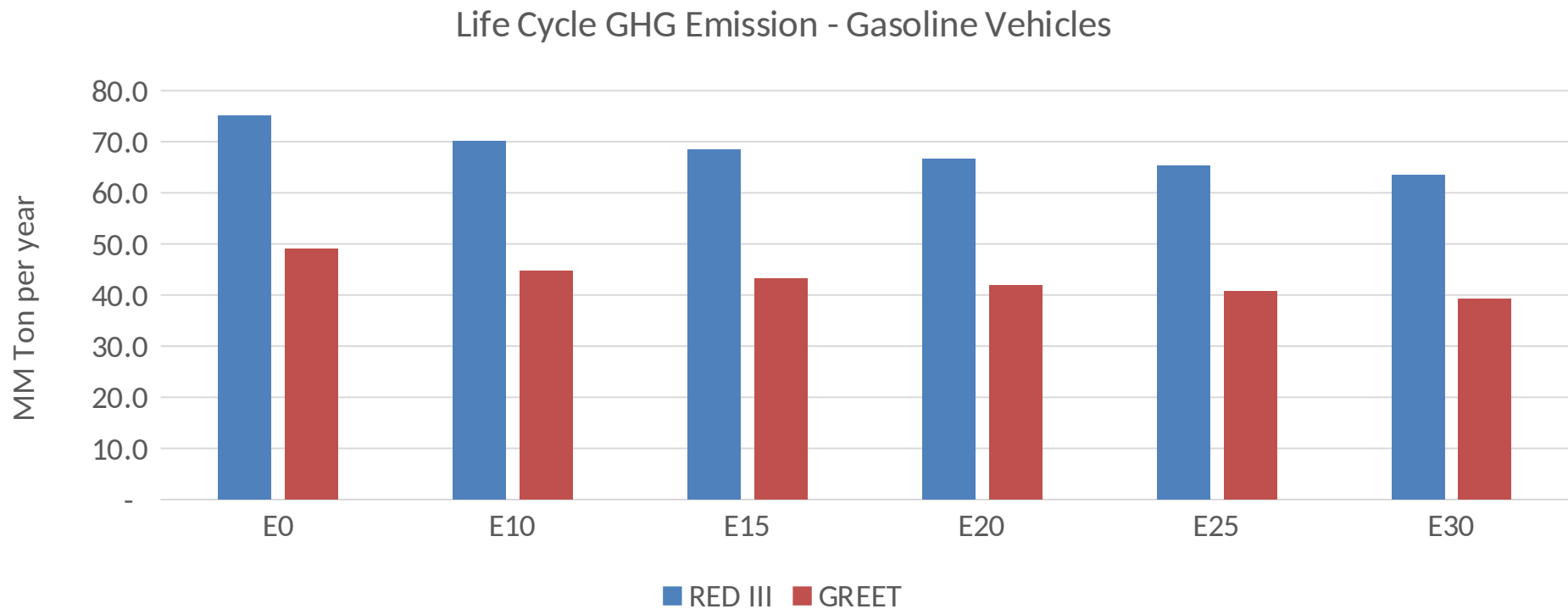
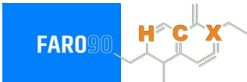
Current GHG Global Emissions Distribution



IPCC Transport Sector Sustainable Development Goals and climate policies Scenario, 2023



Vietnam – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2025 (MMT/año)	489.9
Target @ 2035 (MMT/year)	330.6
Delta	159.3

Source: Greet, RED II/III, climateactiontracker.org, F90